

Foundation architecture - technologies stack

From raw data to graded answer - each block is a component in the source code

DATA (HuggingFace datasets, same org: openlifescienceai)

MedQA-USMLE

HF: openlifescienceai/medqa - eval set (never retrieved)
train 10,178 / val 1,272 / test 1,273

MedMCQA

HF: openlifescienceai/medmcqa - the only retrieval corpus
182,822 solved exam questions (used by V1-V5)

EMBEDDING (embedding model)

qwen3-embedding:4b (V1-V5, 32K)

via Ollama - embeds MedMCQA + every query

VECTOR STORE

ChromaDB - persistent vector store, cosine / HNSW

knowledge_medmcqa_qwen3 (V1-V5, one shared collection)

search_medmcqa (tool)

LONG-TERM MEMORY (cross-episode, from V3)

SqliteStore, persistent (graph/longterm.py)

medqa-lessons (V3)medqa-mistakes (V5)

recall / remember

MULTI-AGENT ORCHESTRATION

LangGraph - StateGraph compiled from a RunConfig, via build_graph(cfg)

config.py - RunConfig / RagConfigroles.py - ROLE_PROMPTS

understandsearch (retrieve+digest)clinical-reasonerdeciderverify V4/V5 onlydistill_mistake V5 only, on a miss

Each role (case-reasoner, evidence-digest, clinical-reasoner, decider, report-verifier, mistake-analyst) is an Agent (agents/base.py): LangChain create_agent, lazily initialized
Reads/writes the shared QASate (short-term memory) + the SqliteStore (long-term memory, same band as ChromaDB)

LLM INFERENCE (shared BaseChatModel)

Ollama (local, default)
qwen2.5:7b

Cloud providers (optional)
anthropic: · openai: · google: · openrouter:

agent.say(...)

EVALUATION

qa/metrics.py - accuracy + 95% CI · invalid-rate · Win/Loss/Tie · McNemar · latency & tokens

AnswerResult(answer, rationale, tokens_in, tokens_out) compared against MedQA-USMLE gold labels

answer(item) → AnswerResult

gold label used for scoring

data

orchestration (LangGraph/LangChain)

embedding / vector (no LLM call)

LLM call

long-term memory (SQLite, no LLM call)

evaluation